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Procedia Computer Science 237 (2024) 645–658

Procedia Computer Science

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International Conference on Industry Sciences and Computer Science Innovation

Blockchain Technology, Structure, and Applications: A Survey

Nazanin Moosavi [a], Hamed Taherdoost [b,c,d,*], Nachaat Mohamed [e], Mitra Madanchian [b,f], Yousef Farhaoui [g], Inam Ullah Khan [h]

[a] *Research and Development Departement, Hamta Business Corporation, Vancouver, Canada*
[b] *Department of Arts, Communications and Social Sciences, University Canada West, Vancouver, Canada*
[c] *QMinded | Quark Minded Technology Inc, Vancouver, Canada*
[d] *College of Technology and Engineering, Westcliff University, Irvine, USA*
[e] *Rabdan Academy, (HLS), Abu Dhabi, UAE*
[f] *Fairleigh Dickinson University, Vancouver, Canada*
[g] *Department of Computer Science, Faculty of Sciences and Technic, Moulay Ismail University, Moroccoo*
[h] *Department of Electronic Engineering, School of Engineering and Applied Science, Isra University, Pakistan*

Abstract

Blockchain-based decentralized technology has gained a lot of attractions and interest in academia and industry. This peer-to-peer and transparent blockchain network makes it an acceptable technology. This decentralized network has many great specifications such as transparency, immutability, irreversibility and auditability. Bitcoin, the first and most successful blockchain application, achieves much success in recent years. This survey article attempts to outline an extensive survey of blockchain technology, investigating its structures, various protocol layers, features, value, challenges, and applications.

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Keywords: Blockchain Technology; Blockchain Structure; Blockchain Features; Blockchain challenges; Blockchain Applications

1. Introduction

Blockchain is proposed as a disruptive and unique technology for realizing distributed ledgers, which gathered enormous research attention in academia and industry. The first introduction of blockchain was from by Satoshi Nakamoto [1] with Bitcoin as a cryptocurrency to provide a trusted peer-to-peer network to perform transactions. Five

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10.1016/j.procs.2024.05.150

generation for blockchain is introduced in scientific communities. Blockchain 1.0 mostly with Bitcoin to propose a trusted network for transactions mostly Bitcoin; Blockchain 2.0 introduces smart contracts which are some programmable rules, Blockchain 3.0 introduces various blockchain use cases in different industries such as telecom, energy, manufacturing, Blockchain 4.0 for real-time business use cases and Blockchain 5.0 for the integration between artificial intelligence (AI) and blockchain [2-4].

Blockchain network consists of some blocks that contain transactional records. Each block contains two main parts, block header that contains some information about the block and block data that contains transactions. Every block header links to the previous block by a cryptographic link. Each transaction contains some details about users who want to the send and receive data and the transferring amount. The sender digitally signed the transaction with a public and private key to ensure security. Sender encrypts the information in each transaction with receiver public key and then receiver decrypt the information with its private key. This is the basic of asymmetric encryption algorithm in blockchain [5].

Blockchain is famous for providing a decentralized network. It deletes the necessity for third parties or authorities to verify and validate the transaction records. Verification is done by using a consensus algorithm to ensure security and trust in a public ledger. The consensus mechanism provide a fair network where every participants can have a merit to append a new block to the block sequences or validate a transaction. The most common consensus mechanism that is although used in Bitcoin network is proof of work (PoW) [6].

Blockchain technology contains many features: transparency which means that all the transactions and process is transparent to all the users in the network, autonomy which makes it a trust-free network by omitting any authority, immutability means that no one can edit or delete the transaction after being initiated, irreversibility which ensures the non-retraceability feature of the technology, and auditability which is the ability that any user can reach the source of a transaction [7].

As the internet has revolutionized information and communication, blockchain is known to revolutionize transactions. With the advent of the internet, various services and applications such as social media, and game streaming were introduced. Blockchain is also proposing many new and interesting services. However, the value captured from the blockchain is different incomparable to the value captured from the internet. Internet captured its values mostly from application layer, whereas blockchain captured its value from protocol layer [8]. Therefore, it is so important to have a general overview of value capturing in blockchain to propose proper and affordable solutions for academic and industrial purposes.

There is hype surrounding most nascent technology. Since the technology is almost new and even experts not know much about it, there are a lot of misunderstandings around it. Blockchain technology has encountered this hype. Although blockchain outlines a new and interesting concept it faces some challenges which needed to be further investigated [9]. Scalability is one big issue of this technology. Two main performance degradations in blockchain efficiency are throughput and validation delay in transactions that are still under investigation to reach satisfaction. There are other limitations such as privacy, energy, regulation, interoperability, and security that require further research to achieve acceptance.

Although blockchain is known to transform the financial market, it can propose novel applications in other sectors. As an example, the integration blockchain technology and internet of things (IoT) can help to overcome security and privacy problems in IoT networks [10, 11]. In addition, multiple healthcare open issues have been tackled with blockchain technology [12-15]. Other verticals such as telecommunication, smart energy, media and entertainment, etc. can gain from this disruptive technology.

The primary contribution of this survey is to offer a deep understanding on blockchain ecosystem. In addition, different specifications such as block structures, different protocol layers consensus protocols, blockchain classification, challenges, the value captured in blockchain, and their application are described in detail. Eventually, this survey article outlines a deep review on blockchain structure and basics and along with blockchain application in different industries that we believe is the strength of this review.

This survey has been structured into seven sections. Section 2 outlines the complete blockchain architectural overview. Section 3 addresses general blockchain features. Section 4 declares the value captured from the blockchain. Section 5 investigates the challenges and issues of this technology. Blockchain application in different industries is explained in section 6 and finally, section 7 depicts the conclusion.

2. Blockchain Architectural Overview

Blockchain network is a string of blocks that holds transactional records through distributed ledgers like public ledgers. In recent years, digitalized ledgers have been used to store data with centralized ownership, but blockchain technology introduces a distributed and decentralized way to store data records. The first block in a block sequence is named genesis block. From the second block onwards, each block holds the hash value of the previous block and hence the sequence of blocks is created.

Every block consists of two parts, block data which is a transaction record stored in each block and block header [12]. The block header consists of some parts as follow:

- Block version- 4 byte: Indicates some rules to validate the block
- Merkle root hash- 32 byte: Hashing method to represent hash of block data
- Previous block hash- 32 byte: Hash value of previous data block
- Timestamp- 4 byte: The time that each block is created since 1970-01-01T00:00 UTC
- nBits- 4 byte: threshold for hashing target value
- Nonce- 4byte: A unique number to create hash for each block data and usually starts with zero

The block data consists of transactions and transaction numerators. An encrypted version of transactions is stored in each block. The encryption of data happens by Merkle tree algorithm. Transactions go into the hash function and then hash value is produced. Next, each of the two-hash value of each transaction forms a pair and go into another hash function to prepare a new hash value. This action continues until reaching one hash value. This procedure is called Merkle tree root hash and is represented in Figure 1.

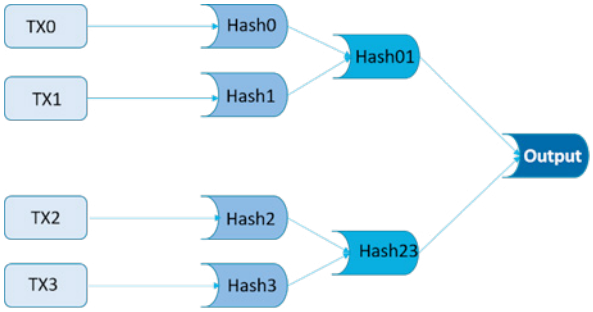
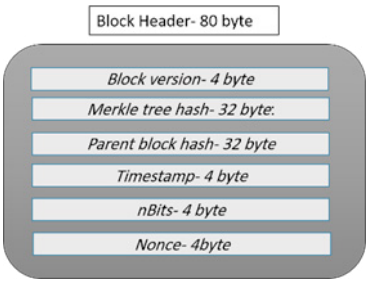
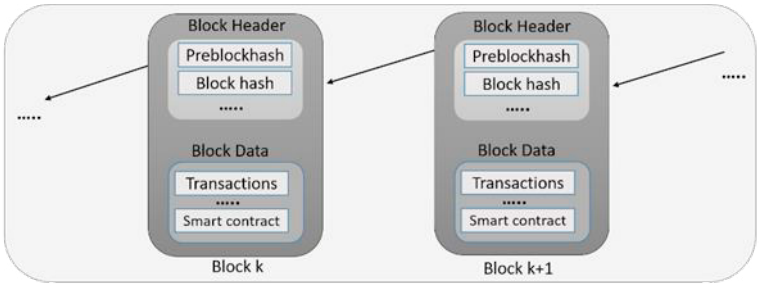


Fig. 1. Merkle Tree Root Hash

Both the transaction size and block size affect the maximum number of transactions in every block. Block Structure Figure 2 shows the pictorial representation of block header and block structure.



a) Block Header



b) Block Structure

Fig. 2. Block Structure and Blockchain Sequence

2.1. How Blockchain Works?

Blockchain users are called nodes. Since the blockchain is a distributed ledger, each node in the network has one copy of the whole blockchain. Nodes are so important since they have an important duty in transactions. There are some nodes with special features that can add a new block to the block sequences. These nodes are miners. Miners perform the process of transaction validation. Once they confirm the transaction, for example in the case of transferring money, the amount of money is transferred from the sender to the receiver [13].

Each node can start a transaction by using a digital signature with public and private key cryptography. Every node knows the public key of other nodes (It is the same as the email address that your contact have it) and the private key is an address that just each node knows (It is the same as the email password). This process is shown in Figure 3. A transaction can be a money transfer between two nodes or any kind of contract between peers in the network. Utilizing the Gossip protocol any transactions are sent to all the nodes. Next, miners verify and validate the transactions and the transaction is appended to the block. This process is called mining. Each miner should expend a lot of computational power to solve a puzzle to release a new block. Winner miner is the first one that solves the puzzle and can create a new block. A transaction fee is also given to the winning miner. Every in the network should verify the new block by employing a consensus protocol which is a set of rules that should be followed during the transactions. After this validation new block is added to the chain [14, 15]. The transaction process presented in Figure 4.

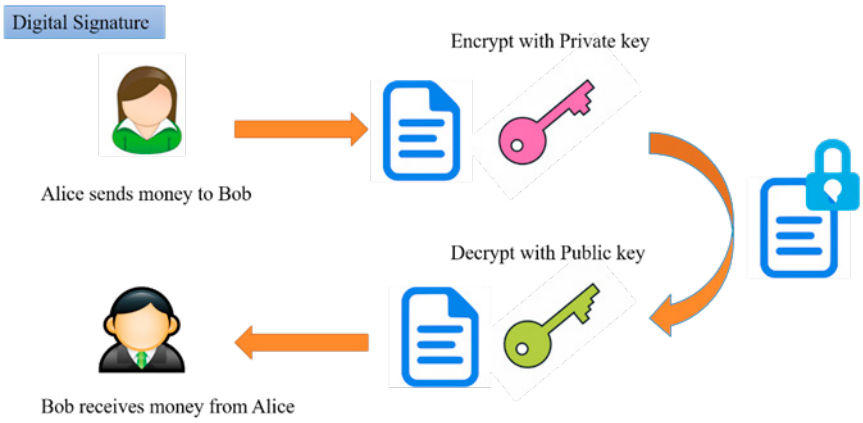


Fig. 3. Digital Signature

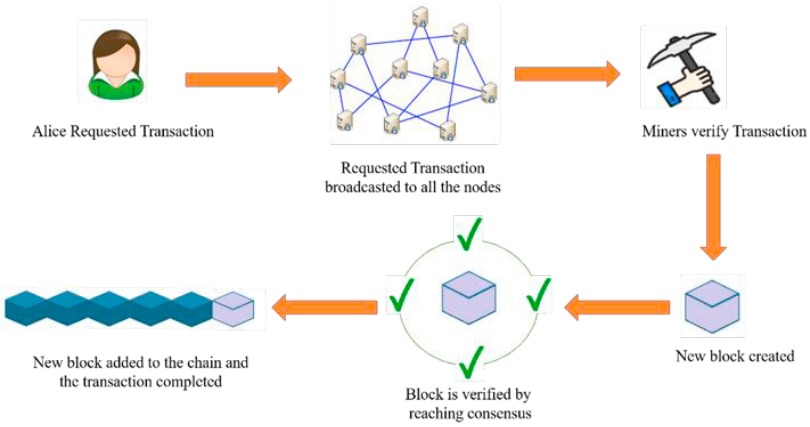


Fig. 4. Functional Diagram of Transactional Flow in Blockchain

Later in this section, 2.2 introduces different blockchain classification, 2.3 discusses blockchain protocol layers, 2.4 explains some consensus protocols, 2.5 defines smart contract and 2.6 outlines different generation of blockchain.

2.2. Blockchain Classification

Blockchain networks are classified into two models based on the permission for appending a new block. If any user can access to add a new block, the blockchain is permissionless. But if only a few specified users can add a new block to the network the blockchain is permitted. Simply, a permissionless blockchain is the same as public internet that anyone can access to it, but a permissioned blockchain is the same as an Intranet that is under control. Permissioned blockchain is often used by groups or organizations [7]. Blockchain classification is proposed in Figure 5.

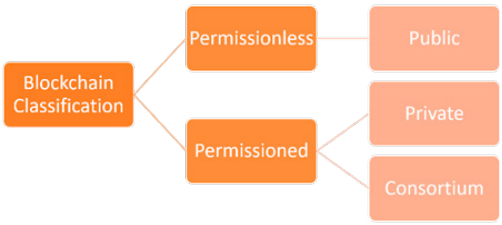


Fig. 5. Blockchain Classification

2.2.1. Permissionless Blockchain

One good example of permissionless classification is public blockchain [16]. In a public blockchain, any user can enter or exit the chain at any time. All the users have one copy of all the transactions and since the transactions are visible to anybody no one can change it. The most famous public blockchain is Bitcoin [17, 18].

2.2.2. Permissioned Blockchain

Permissioned blockchain is categorized into two main groups: private blockchain and consortium blockchain. In the first group, just a selection of nodes can access the network and validate the transaction. This limitation makes the transaction to be faster and even if the blockchain grows, the transaction time is efficient. In this type of blockchain, a completely secure transaction can happen. The main disadvantage of private blockchain is that since it is under control of one organization, the procedure of validation becomes centralized [18].

A consortium blockchain has some specification of public and private blockchain where there is more than one organization that is controlling the network [19]. Here is the comparison between different blockchain classifications in Table 1.

Table 1. Comparison between blockchain classifications

Category of blockchain	Participant	Consensus protocols	Scalability	Efficiency	Energy Consumption	Example
Public	Anonymous	PoW/PoS	High	Low	High	Bitcoin Ethereum
Private	Identified	PBFT	Low	High	Low	Multichan/Blockstack
Consortium	Identified	PBFT	Low	High	Low	Ripple/Hyperledger

2.3. Blockchain Protocol Layers

The blockchain framework has three layers called network layer, data layer, and application layer as shown in Figure 4. Data layer responsibility is building new blocks, the network layer is responsible for interaction protocols between users and environments and finally, the application layer represents the various application of this technology [20, 21]. Further details on these layers are given below.

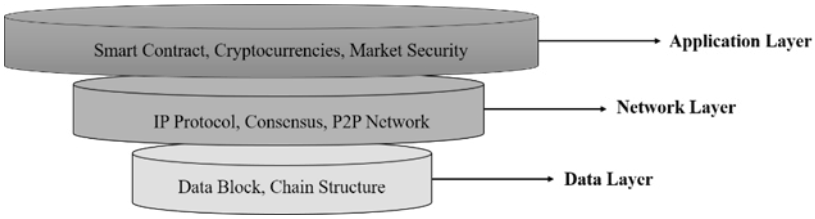


Fig. 6. Blockchain Layers

2.3.1. Data Layer

Data layer provides the functional and procedural means to make data records (blocks) in blockchain ledger. This layer is concerned with the data structures and algorithms such as digital signature, Merkle tree, timestamp, hash pointer, etc. It makes the blockchain network decentralized, transparent and persistent.

2.3.2. Network layer

Network layer provides the means of decentralizing the network via IP protocols and peer-to-peer networks. The consensus algorithm and its implementation that is used for distributed agreement for block validation is outlined in this layer.

2.3.3. Application Layer

This layer represents the various applications of blockchain in different sectors to leverage its distributed and decentralized features. Although financial market and cryptocurrencies are the most popular application of blockchain, but it can be applicable in many other verticals like healthcare, energy and agriculture to meet the growing demand for security and privacy of users. The application layer will be more detailed in further sections.

2.4. Consensus Protocols

The idea behind the consensus protocols is to give a right to a node which provides sufficient effort and computational power to add a new block to blockchain. Generally, this protocol is resource wasteful, although there are many research on proposing a proper consensus protocol to increase the blockchain efficiency and to encourage application requirements. Some of most important consensus protocols are defined here [16].

2.4.1. Proof of Work (PoW)

PoW is the famous consensus protocol among blockchain networks. The goal of this protocol is to find the solution for a mathematical puzzle and then as a reward the winner node that investigates more computing power can append a new block to the block sequence. Blockchain users try to find the hash value that has the pre-defined condition. Miners use nonce in order to find the hash value. When one miner succeeds to find the nonce, it publish it to all the nodes, then other miners should verify and validate it and then append it to the blockchain. Although this process is somehow not so hard, a major computing resources is needed, which is not environmentally friendly. Hence, other consensus protocols have been studied [17].

2.4.2. Proof of Stack (PoS)

PoS does not need a high amount of computing power. Hence, it is referred to as an alternative to PoW from an energy saving point of view. In this protocol, there are no miners. Instead, there are validators that based on their stack can participate in finding the new block. Each validator bets on a block by using its stack. Then if the block is approved, it gathers the transaction fees in the block. In this way by consuming less energy, less latency and better throughput are achieved [18].

2.4.3. Practical Byzantine Fault Tolerance (PBFT)

Byzantine fault tolerance (BFT) is a situation when two nodes, in the presence of some malicious nodes, want to communicate with each other. PBFT is one example of BFT that can withstand Byzantine fault. There is one node called a leader in this network and the others called backups. The protocol has three phases. In the first phase namely *Preprepared*, the leader publishes the intended value to be committed to the blockchain. In the second phase, namely *Prepared*, backup nodes intend value to be committed. In the third phase namely *Commit*, if more than two-thirds of backups agree on the value in the Prepare phase, the value would be committed. Since this protocol requires several rounds to be completed, it is not a scalable solution [19]. Various consensus protocols have different characteristics. **Erro! A origem da referência não foi encontrada.** proposes a comparison between previous protocols.

Table 2. Comparison between different Consensus Protocols

Consensus Algorithm	**latency**	**Number of Transaction**	**Scalability**	**Block creation speed**	**Energy Consumption**	**Use Cases**
Proof of-Work (PoW)	6-60 min	10s	Strong	Slow	High	Bitcoin Ethereum
Proof of-Stake (PoS)	10-60 min	10s	Strong	Fast	Low	Peercoin
Practical Byzantine Fault Tolerance(BFT)	1-60 sec	1000s	Weak	Fast	Very Low	Hyperledger

2.5. Smart Contract

N. Szabo first introduced a smart contract in 1997 [25]. The idea of a smart contract is to predefine some roles, which are programmable codes between two participants. The concept of a smart contract is just like a cryptographic box. It can be opened when some conditions are met. There are no third party or intermediaries in a smart contract hence it is a self-executing code. To be specific if ‘x’ happens, then ‘y’ applies. For ease of understanding, Table 3 compares traditional contracts with smart contracts.

Table 3. Comparison between smart contract and traditional contract

Requirements	**Smart Contract**	**Traditional contract**
Third-party	Not required	Banks, Regulators
Execution time	Minutes	Days
Transparency	Available	Not available
Signature	Digital	Manual
Process	Automatic	Manual
Achieve	Easy	Hard

Security	Highly secure	Limited

Conditions and terms are written with programmable codes and are implemented on the blockchain platform. Based on these codes some assets, for example, money, shares, or documents are held in the smart contract until the contract terms are enforced and then triggered automatically [20]. The characteristic of the smart contract makes the way to empower distributed electronic virtual machines (EVM) as an example Ethereum virtual machine [21]. The execution of the smart contract is applied to the node in EVM for progress. Since all the users in the network have the same blockchain, they have the same results by executing the EVM code when one smart contract applies. All the results are stored in the blockchain ledger.

However, what happens if the conditions are not met? In this condition, the smart contract will be executed automatically, and network users will gain compensation. This is the main advantage of the smart contract. With no third parties or manual process, compensation is given to the users [22].

2.6. Blockchain Generations

Blockchain evolution from its first day until today is described in the Figure 7 [28-30].

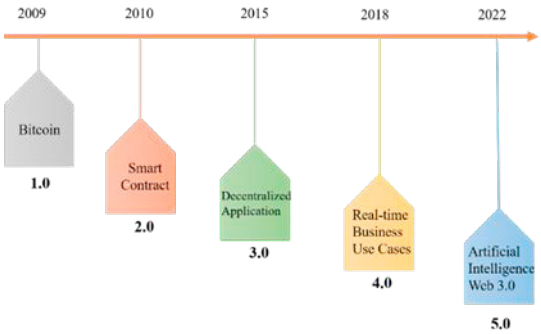


Fig. 7. Blockchain Generation

2.6.1. Blockchain 1.0

The first generation of Blockchain was introduced to the world with Bitcoin, which is the most famous digital currency and the first famous application of blockchain technology. The idea behind blockchain 1.0 is virtual cryptocurrencies, financial services, and payments.

2.6.2. Blockchain 2.0

The second generation of blockchain technology introduces programmable platforms and applications in different sectors. Smart contracts and public ledgers to keep record was first introduced in this generation. Blockchain 2.0 proposed some other applications to financial systems beyond cryptocurrencies such as smart property, payment clearing, and banking instruments.

2.6.3. Blockchain 3.0

The application of blockchain in different sectors such as healthcare, government, industry, technology, and art is proposed in blockchain 3.0. Improving smart contracts for various decentralized applications is introduced at this level.

2.6.4. Blockchain 4.0

The main target of blockchain 4.0 is to focus on business applications in real-time. Public ledger and distributed database in real-time has paved the way for seamless application in industry 4 which mitigates the use of paper-based regulation by using smart contract and consensus algorithms.

2.6.5. Blockchain 5.0

The goal of Blockchain 5.0 is to focus on the integration of artificial intelligence (AI) and distributed ledger technology (DLT) in order to develop the decentralizes Web 3.0 applications. Its main concentrate is on

data privacy, security, and interoperability [3].

3. Blockchain Features

Six main features of blockchain, which made it one of the technology trends these days, is presented in **Erro! A origem da referência não foi encontrada.**

3.1. Decentralized

Decentralization is the most important feature of blockchain technology. It omits the need for a third party or centralized authority to validate the transaction. With this feature, every node in the network can transact with each other without having any information about their real identity or their location. Moreover, since there is no centralized system, every node has the entire information of the network, and the data is stored in multiple systems. When a new block is added to the chain, the data is updated in all the systems [23].

3.2. Transparent

All the algorithms and transactions in blockchain technology are transparent to all the users in the network. It leads the way for accountability and integrity of data in the network since no one can change, add or delete data. This feature is unprecedented for the financial systems, which makes the blockchain technology popular in the financial market [24].

3.3. Autonomy

A Blockchain network is called a *trust-free* network, which omits the need for an authority to ensure trust. In this system, trust is not an issue, and guarantees trust for all network transactions. Cryptographically algorithms in the blockchain network replace the governor's authority to ensure trust [25].

3.4. Immutable

The immutability characteristic of blockchain means once a transaction happens and the data is stored in the blockchain sequence; no one can edit or delete it. After transaction registration, they can just be validated and verified. This characteristic makes a blockchain immutable and constant. Since transactions are stored in all the nodes, if any modification happened to one transaction, all the nodes are informed and try to fix it [26].

3.5. Irreversible

This feature proposes that once a transaction is saved in the network, it cannot be retraced. This happens due to the usage of the hash function that gives an encrypted version of the data in each data block of the blockchain. Since the hash function is a one-way function, the input cannot be retraced from the hash data. In addition, the hash function is a deterministic function which means that just the same output, has the same input. Hence, just a small change in inputs, causes different output. Both two properties of the hash function guarantee the irreversibility feature of blockchain [27].

3.6. Auditability

Each transaction is recorded in a distributed ledger and with a timestamp that is stored in a block header. Hence, one can detect the previous nodes by accessing any node in the network [28]. It is a beneficial feature of blockchain

in some applications since it is so important to know the source of the data for example the source of a huge transaction or money transfer in financial services to avoid fraud or money laundering [23].

4. Value Captured in Blockchain

Blockchain is known to be a transformational technology since it provides an open-source basis for security services in various sectors and financial markets. Some visionaries simulate blockchain with the internet. In the early days of the internet, no one could guess how the internet was going to influence our lives. From smartphones to diverse applications for example social media, video conferences, game streaming, etc. Due to the fast and wide development in blockchain applications, some specialists envisage blockchain would transform transactions similar to the way that the internet has transformed information and communication technology [24, 29, 30].

The main difference between the value captured on the internet and blockchain is the protocol layer. To investigate this difference, it should be noted that the internet runs on open protocols such as TCP, IP, HTTP, etc. These protocols are usually preserved by non-profit organizations. These protocols produce different applications such as Facebook, Gmail, and Wikipedia, which people use to communicate with each other or get information about different things. This means that the value is captured from different applications. Hence the value captured from the internet is mainly from the application layer, not the protocol layer and therefore it is composed of thin protocol and fat applications [8].

Next, we want to investigate the value captured in the blockchain. Blockchain consists of fat protocols and thin applications. Value is mainly captured from the protocol layer and just a small amount of it is captured from the application layer. As an example, the market capitalization of Ethereum [31], which is the most famous blockchain network after Bitcoin, until 2021 is $528 billion, however applications that run on top of this network get a small fraction of its value. Figure 8 proposes a comparison between internet and blockchain from the perspective of value captured.

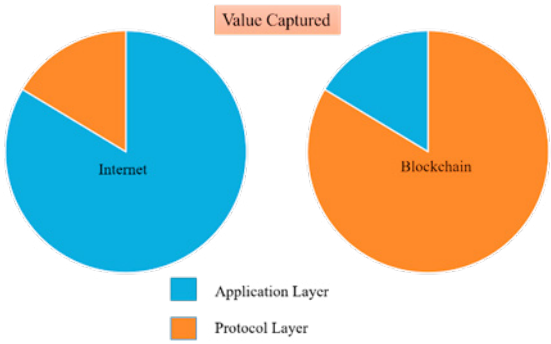


Fig. 8. Value captured comparison between Internet and Blockchain

5. Blockchain issues and challenges

Nowadays, blockchain technology is a new trend in both academia and industry. With its special features, it is known to revolutionize the financial market. However, it should be considered that blockchain technology is still in its infancy time and it suffers from some limitations which make it not acceptable in some industries [32, 33]. In this section, challenges of blockchain technology are investigated. **Erro! A origem da referência não foi encontrada.**9 shows some challenges of blockchain technology.

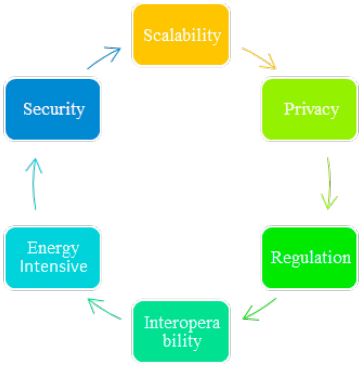


Fig. 9. Blockchain issues and challengesScalability

With the popularity of blockchain application in various sectors, the number of transactions is increasing diurnally. However, there is always a concern about whether blockchain can handle this large amount of transactions or not. Hence, the scalability of the network is of high importance and many researchers have started working on it. By increasing the number of replicas, two important performance features change, throughput which depicts how many transactions can happen in a second and latency which depicts the time delay for appending a block to the blockchain [34]. As an example, Bitcoin has a very low throughput which can only process 6-10 transactions per second. However, in comparison with Bitcoin Ethereum can process 15 transactions [35], Ripple can process 1500 transactions [36] and Visa can process more than 2000 transaction per second[37]. Hence, Bitcoin is not a good solution for real-time transactions in some industries. In addition, Bitcoin network needs around 10 minutes to append a new block to the network. Although Bitcoin and PoW can somehow assure scalability it suffers from low throughput, high latency, and a large amount of processing power.

It should be noted that since the capacity of the block is so small (block capacity of Bitcoin is 1MB), miners are eager to mine the blocks with a high transaction fee, hence small transactions might be delayed. Although this characteristic ensures the security of the network it slows down the transactions process compared with existing systems.

The DCS (decentralized, consistent and scalable) theorem which is shown in **Erro! A origem da referência não foi encontrada.**0 can propose some ideas around the scalability issues on blockchain [38]. According to this theorem, a blockchain network can not have all the three DSC features simultaneously and it can meet only two of the features. Albeit, it gives low throughput and high latency, e.g., low volume and slow transaction.

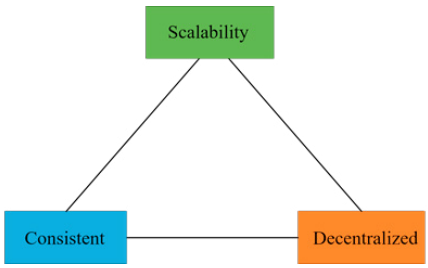


Fig. 10. DCS Theorem

5.1. Privacy

Each user in blockchain can initiate the transaction without its real identity and with user pseudonyms, which provides safety and privacy for the user's personal information. However, some researchers studied privacy leakages in the blockchain network. Researches in [39] showed that since the value of transaction and its balance is visible to everyone with public key, the blockchain network is vulnerable. Authors in [40] provide a method that links the user's pseudonyms to IP addresses even if users are behind firewalls. In addition works in [41] reveal a way to show the

user's identity by using transaction history. Consequently, the privacy and safety of users should be considered in the initial stage of building a blockchain network based on the level of privacy that each sector or organization needed.

5.2. Regulation

Based on discussions in [42] unclear regulations and lack of standardization are two factors to limit the blockchain's popularity in different industries. However, the principle of blockchain and cryptocurrencies is to create an open and trusted platform for trading, this open network eases the presence of malicious attackers. Hence, government and developers should study some regulations to create cryptocurrency ecosystem, which is legal and ethical [43]. Regulation and rules could affect the growth of technology in the fastest time.

Researches in [44] agreed that if blockchain wants to replace existing solutions, it should overcome uncertain regulation, cultural issues, and integration issues. Nowadays there are no complete regulations and rules for cryptocurrencies, which makes the usage of the blockchain network, a big challenge in the financial markets.

5.3. Interoperability

Interoperability refers to an ability of different software to collaborate, connect and share data/information [45]. Although various industries are interested in using blockchain, there are no standard rules and protocols to allow them to collaborate. This lack of interoperability has attenuated the development of the blockchain ecosystem. Hence, emphasizing cryptocurrency as the main blockchain platform is a better solution instead of offering a different solution for a variety of industries. Although, the lack of interoperability between current internal platforms and blockchain solution cause some challenges to the existing companies [46]. In addition to these issues, active blockchain networks developed on different platforms with different programming languages, consensus protocols, and privacy features make interoperability harder.

5.4. Energy Intensive

PoW with its Bitcoin evolution is the most important consensus protocol in the blockchain network. However, its strong desire to use a huge amount of energy for computation causes a lot of environmental problems [47]. One way to understand this huge amount of energy is to compare it with the amount of energy that payment systems such as VISA need for their transaction. In [48], it is shown that the amount of energy for 1 bitcoin transaction is even twice the amount of energy for 100,000 VISA transactions. Although blockchain is proposed to remove the need for central authorities in the financial market, this huge amount of energy is too high to bear. Consequently, new protocols such as PoS, which use less energy, should be considered as alternatives.

5.5. Security

Blockchain has shown promising applications in financial market. However, users are worried about its security [49]. In addition, smart contracts suffers from security risks, which may produces loss [50]. As an example by an attack to DAO smart contract, the attackers stole around 60 million dollars by applying a recursive calling vulnerability [51]. Two main security vulnerabilities and attacks are as follow.

5.5.1. Majority attacks

Most attacks (51%) happen when more than half of computing power is under the control of one user or one group of users who have established a mining pool. In a 51% attack, attackers can forbid other users to affirm the blocks and they can inverse the transactions. Attackers can also hijack the entire process of writing blocks, which introduces wrong blocks. In addition, since the network is under the control of attackers they can revise the history of the transactions or falsifies it [19, 52]. Double spending is another attack that happens when miners own more than half of the computing power. It is a transaction attack that happens during the verification process. By producing fake transactions, miners spend the same digital currency twice and transactions cannot be validated. However, this attack is not able to build a new account, grab funds or create false transactions [53].

5.5.2. Selfish mining

A selfish mining strategy happens, when some selfish miners do not broadcast their new blocks after mining, and hence a private branch is produced. By this way, the private Blockchain becomes longer than the current public chain, all the miners admit it. In this situation, the honest miners waste their computing power on the useless chain and selfish miners have no competitors to mine new blocks. Consequently, selfish miners get more rewards with less effort. This attack can be prevented by adding trusted miners [54].

6. Conclusion

In this work, a detailed survey on blockchain technology is presented. Blockchain technology by enabling decentralized networks provides an immutable, tamper-proof, and secure network by eliminating central authorities. This survey explains the complete details of block structures with the mining method to append a new block to the blockchain. Blockchain generation and classification, different protocol layers, and the definition of the smart contract is been fully investigated. In addition, this paper investigates main blockchain technology specifications such as decentralization, immutability, transparency, etc. Although blockchain is known to be such as the internet, which is going to transform the transaction, the way that it captures value, is much different from the internet.

Blockchain value mostly comes from the protocol layer instead of the application layer. Subsequently, it is so important to know the meaning of fat protocol and thin application. Next blockchain challenges and issues are outlined. Eventually, the application of blockchain in different verticals and industries is elucidated. In this comprehensive survey, we offer a brief but complete outline of blockchain with its structure, principles, protocol layers, consensus mechanisms, classification, usage in industries and we provide a concise but extensive review of blockchain with its background, principles, consensus mechanisms, classifications, applications, and challenges inside a one survey valuable for a blockchain innovation devotees. The principles and discoveries discussed here can be examined profoundly based on the needs of academia circles and industries.

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